Hospital Management System (HOMS)

1. Introduction

Hospital Management System (HOMS) is a software system used to manage hospital operations such as patient registration, doctor scheduling, appointment booking, treatment, admission, and billing.

In traditional systems, these tasks are done manually which leads to delays and errors. This system helps to automate the process, improve efficiency, and maintain accurate records.

The main focus of this system is **patient-doctor interaction**, where patients can easily consult doctors and receive proper treatment.

2. Objectives

- To automate hospital activities
- To manage patient and doctor information
- To simplify appointment booking
- To store medical records properly
- To generate bills automatically
- To improve service quality and reduce errors

3. Functional Requirements

The system should be able to:

- Register new patients
- Allow patient login and profile management
- Show available doctors and their specialization
- Book, cancel, and reschedule appointments
- Allow doctors to view appointments
- Record diagnosis and prescription
- Admit patients when necessary
- Allocate rooms to patients
- Generate bills automatically
- Accept and record payments

4. Non-Functional Requirements

The system should:

- Provide fast response time
- Ensure data security and privacy
- Be user-friendly and easy to use
- Maintain data accuracy and consistency
- Support backup and recovery
- Be reliable and scalable

5. Use Case Modeling

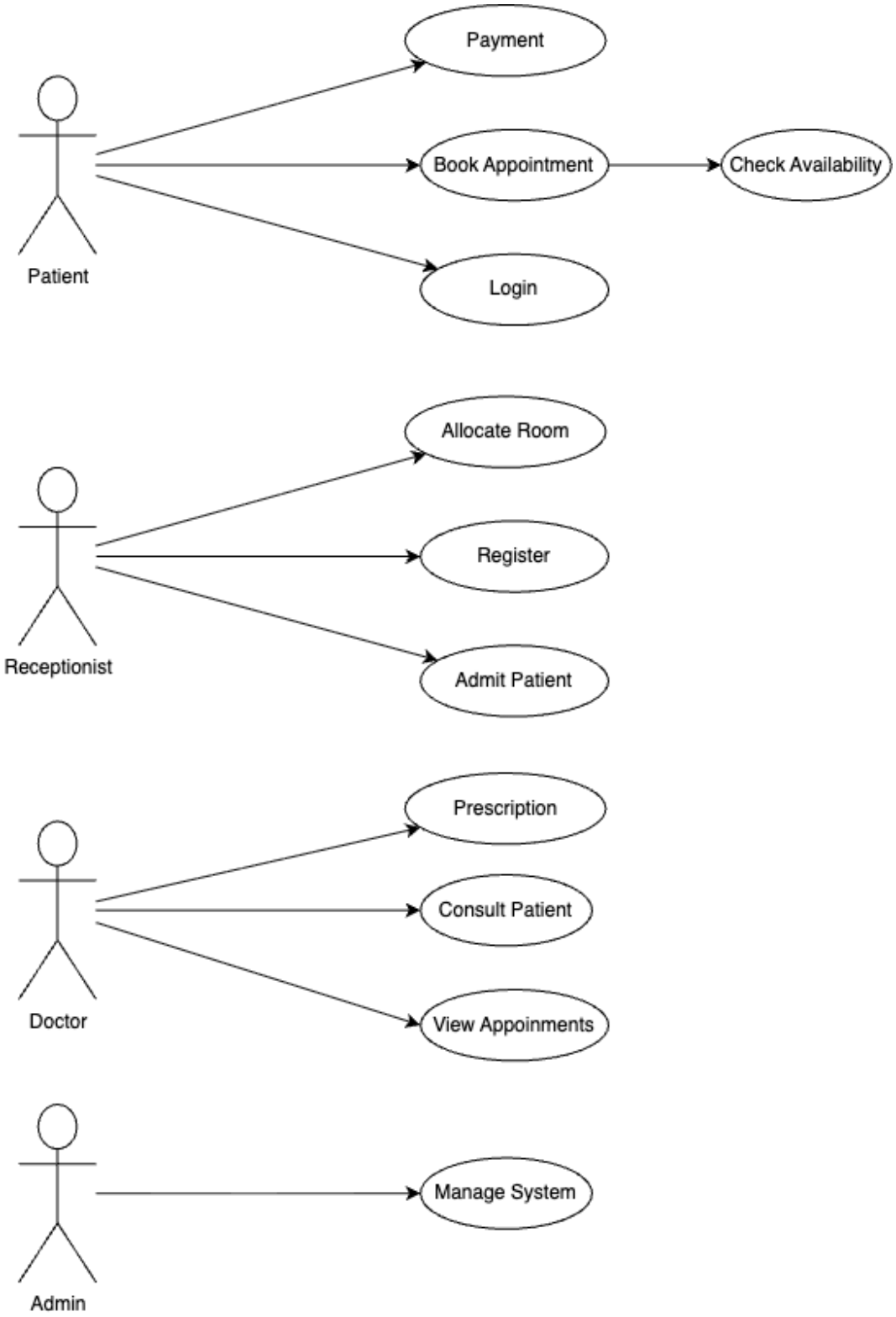
Actors:

- Patient
- Doctor
- Receptionist
- Admin

Main Use Cases:

- Register Patient
- Login
- Book Appointment
- Check Doctor Availability
- Consult Patient
- Create Prescription
- Admit Patient
- Allocate Room
- Generate Bill
- Make Payment

Use Case Diagram:



6. Structural Design (Class Diagram)

Main Classes:

- User
- Patient
- Doctor
- Appointment
- MedicalRecord
- Prescription
- Room
- Bill

Description:

- **User** is the parent class
- **Patient** and **Doctor** inherit from User
- Patient books appointments
- Doctor handles appointments
- MedicalRecord stores patient history
- Bill stores payment details

Attributes and Methods

User

- Attributes: id, name
- Method: login()

Patient

- Method: bookAppointment()

Doctor

- Method: diagnose()

Appointment

- Attributes: date, status

MedicalRecord

- Attribute: diagnosis

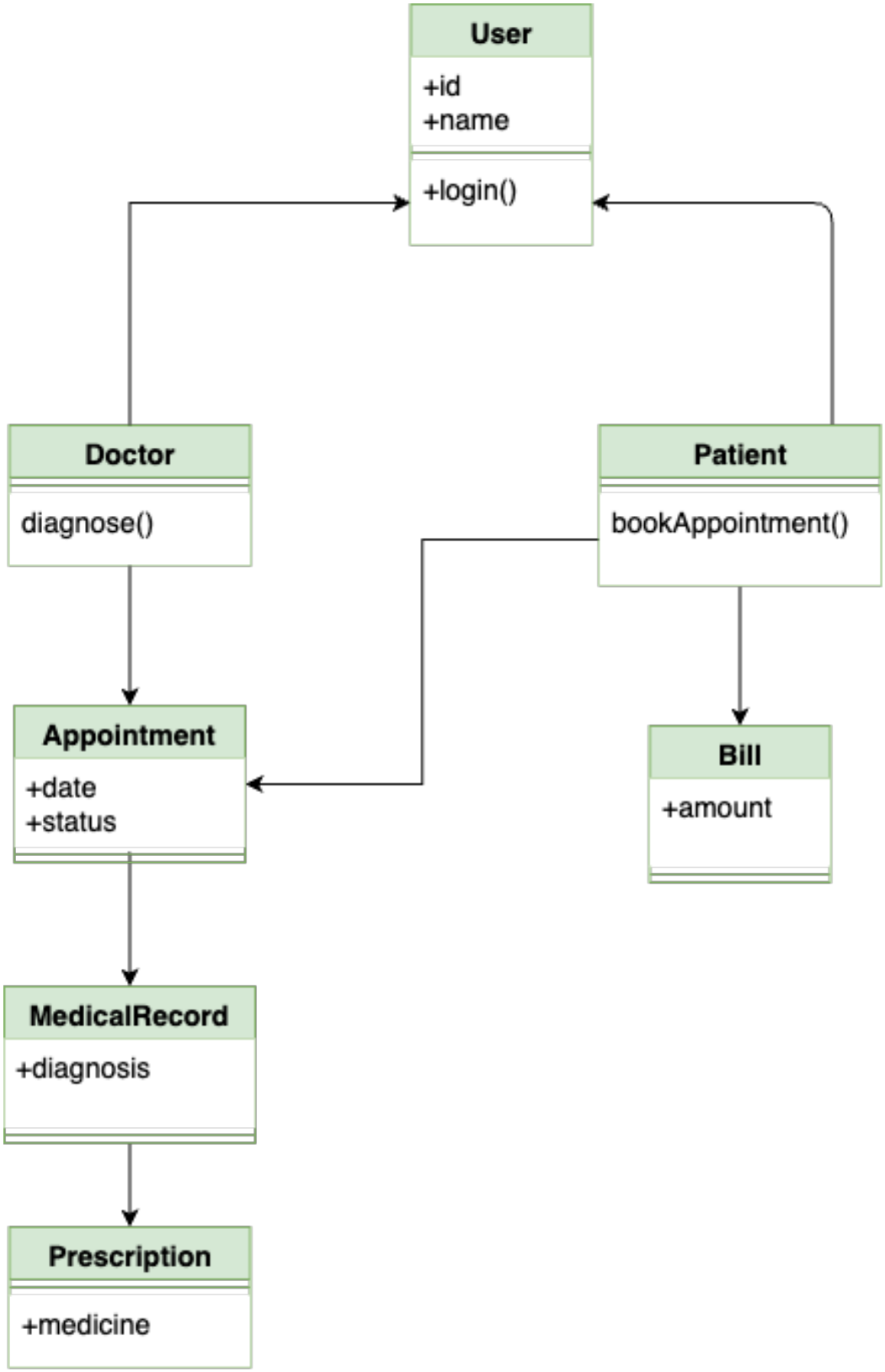
Prescription

- Attribute: medicine

Bill

- Attribute: amount

Class Diagram



7. Database Design (ER Diagram)

7.1 Entities

- Patient
- Doctor
- Appointment
- Medical_Record
- Prescription
- Bill

7.2 Relationships

- Patient books Appointment
- Doctor attends Appointment
- Appointment creates Medical Record
- Medical Record generates Prescription
- Patient receives Bill

7.3 Suggested Primary Keys and Foreign Keys

PATIENT

- patient_id (PK)
- name
- age
- gender
- phone

DOCTOR

- doctor_id (PK)
- name
- specialization
- phone

APPOINTMENT

- appointment_id (PK)
- patient_id (FK)
- doctor_id (FK)
- date
- status

MEDICAL_RECORD

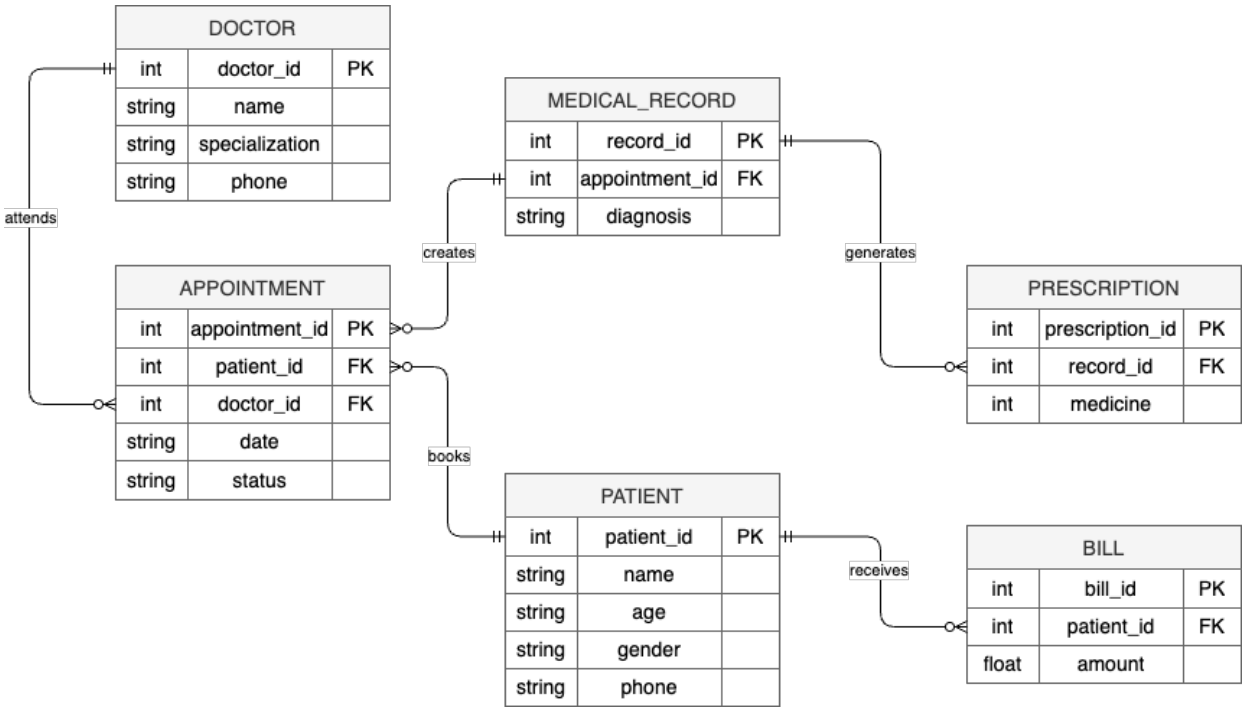
- record_id (PK)
- appointment_id (FK)
- diagnosis

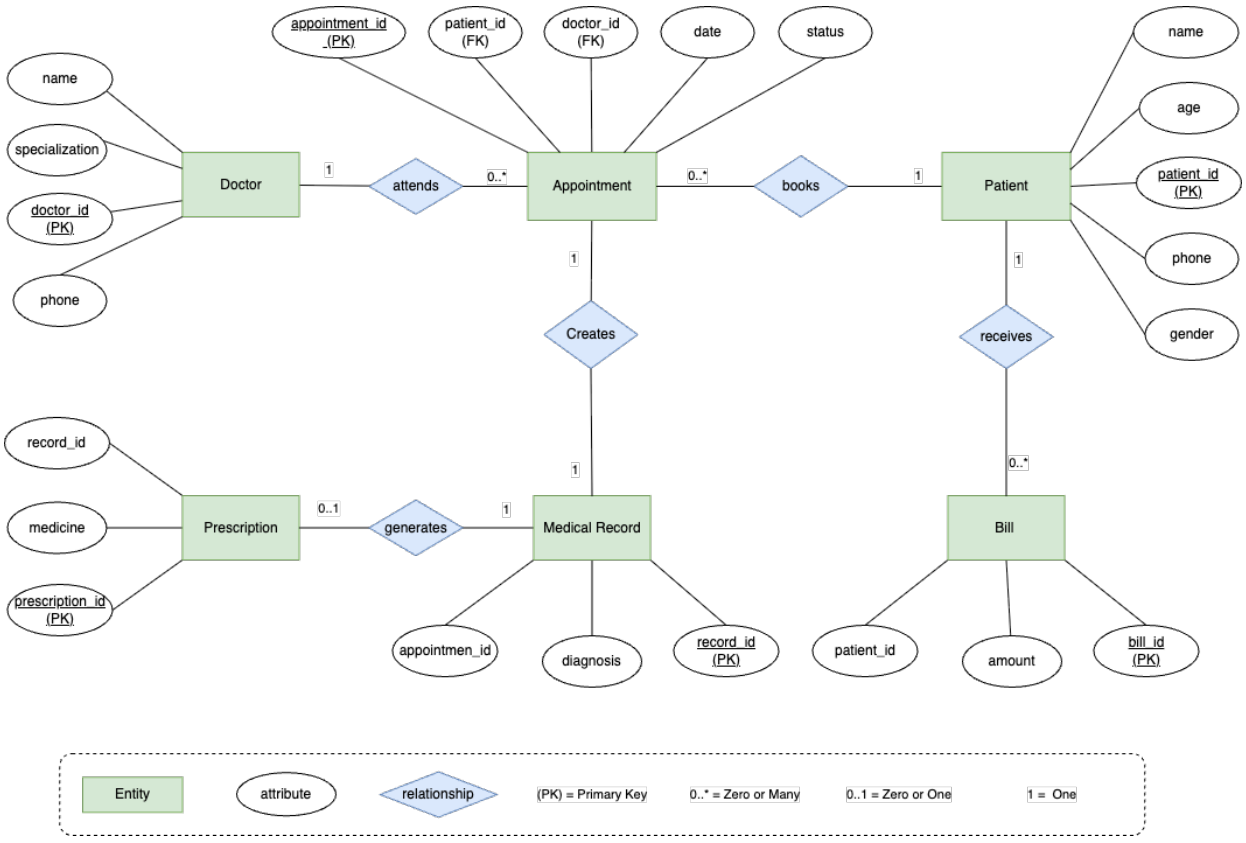
PRESCRIPTION

- prescription_id (PK)
- record_id (FK)
- medicine

BILL

- bill_id (PK)
- patient_id (FK)
- amount





8. Advantages

- Reduces manual work
- Saves time
- Improves data accuracy
- Easy record management
- Better patient service

9. Limitations

- Requires technical knowledge
- System failure can affect work
- Initial setup cost is high

10. Conclusion

The Hospital Management System is an efficient solution for managing hospital activities. It improves patient-doctor interaction, reduces errors, and enhances overall hospital performance. The system ensures better service and organized data management.

11. Simple Web Implementation

index.html

```html
<!DOCTYPE html>
<html>
<head>
  <title>Hospital Management System</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>

<header class="hero">
  <h1>Hospital Management System</h1>
  <p>Simple Offline Portal</p>
</header>

<nav>
  <a href="#patients">Patient</a>
  <a href="#doctors">Doctors</a>
  <a href="#appointment">Appointment</a>
  <a href="#records">Records</a>
```

```html
</nav>

<div class="container">

<!-- PATIENT -->
<section id="patients" class="card">
  <h2>Register Patient</h2>
  <form id="patientForm">
    <input type="text" id="pname" placeholder="Name" required>
    <input type="number" id="page" placeholder="Age" required>
    <input type="text" id="pphone" placeholder="Phone" required>
    <button type="submit">Add Patient</button>
  </form>
</section>

<!-- DOCTORS -->
<section id="doctors" class="card">
  <h2>Doctors</h2>
  <div id="doctorList"></div>
</section>

<!-- APPOINTMENT -->
<section id="appointment" class="card">
  <h2>Book Appointment</h2>
  <form id="appointmentForm">
    <input type="text" id="patientId" placeholder="Patient Name" required>
    <input type="text" id="doctorId" placeholder="Doctor ID" required>
    <input type="date" id="date" required>
    <button type="submit">Book</button>
  </form>
</section>

<!-- RECORDS -->
<section id="records" class="card">
  <div class="section-title">
    <h2>Appointments</h2>
    <button onclick="clearAllAppointments()" class="danger-btn">Clear
All</button>
  </div>
  <div id="recordsList"></div>
</section>
</div>

<script src="script.js"></script>
</body>
</html>
```

styles.css

```
body {
  font-family: Arial;
  background: #f4f7fb;
}

.hero {
  background: #1d4ed8;
  color: white;
  padding: 20px;
  text-align: center;
}

nav {
  background: white;
  padding: 10px;
  text-align: center;
}

nav a {
  margin: 10px;
  text-decoration: none;
  color: #1d4ed8;
}

.container {
  width: 80%;
  margin: auto;
}

.card {
  background: white;
  padding: 20px;
  margin: 20px 0;
  border-radius: 10px;
}

input {
  display: block;
  margin: 10px 0;
  padding: 10px;
  width: 100%;
}

button {
  padding: 10px;
  background: #1d4ed8;
  color: white;
  border: none;
}
```

```css
.section-title {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.danger-btn {
  background: #dc2626;
}

.danger-btn:hover {
  background: #b91c1c;
}

.cancel-btn {
  background: #f97316;
  margin-top: 8px;
}

.cancel-btn:hover {
  background: #ea580c;
}

.complete-btn {
  background: #16a34a;
  margin-top: 8px;
  margin-right: 8px;
}

.complete-btn:hover {
  background: #15803d;
}
```

script.js

```js
const doctorList = document.getElementById("doctorList");
const recordsList = document.getElementById("recordsList");

// LOAD DOCTORS
fetch("doctors.txt")
  .then(res => res.text())
  .then(data => {
    const doctors = data.split("\n");
    doctorList.innerHTML = doctors.map(d => {
      const [id, name, spec] = d.split("|");
      return `<p><b>${id}</b> - ${name} (${spec})</p>`;
    }).join("");
  });
```

```javascript
// LOCAL STORAGE
function getAppointments() {
  return JSON.parse(localStorage.getItem("appointments")) || [];
}

function saveAppointments(data) {
  localStorage.setItem("appointments", JSON.stringify(data));
}

// PATIENT REGISTER
document.getElementById("patientForm").addEventListener("submit", e => {
  e.preventDefault();
  alert("Patient Registered");
});

// APPOINTMENT
document.getElementById("appointmentForm").addEventListener("submit", e => {
  e.preventDefault();

  const patient = document.getElementById("patientId").value;
  const doctor = document.getElementById("doctorId").value;
  const date = document.getElementById("date").value;

  let data = getAppointments();

  data.push({
    patient,
    doctor,
    date,
    status: "Pending",
    bill: 500
  });

  saveAppointments(data);
  render();

  alert("Appointment Booked");
});

// RENDER
function render() {
  const data = getAppointments();

  if (data.length === 0) {
    recordsList.innerHTML = "<p>No appointments found.</p>";
    return;
  }

  recordsList.innerHTML = data.map((d, index) => `
    <div class="card">
```

```
      <p><b>Patient:</b> ${d.patient}</p>
      <p><b>Doctor:</b> ${d.doctor}</p>
      <p><b>Date:</b> ${d.date}</p>
      <p><b>Bill:</b> ${d.bill} BDT</p>
      <p><b>Status:</b> ${d.status}</p>

      ${d.status === "Pending" ? `
        <button class="complete-btn" onclick="completeAppointment(${index})">
          Complete Appointment
        </button>

        <button class="cancel-btn" onclick="cancelAppointment(${index})">
          Cancel Appointment
        </button>
      ` : ""}
    </div>
  `).join("");
}

function completeAppointment(index) {
  const data = getAppointments();

  data[index].status = "Completed";
  data[index].bill = 500;

  saveAppointments(data);
  render();

  alert("Appointment completed.");
}

function cancelAppointment(index) {
  const data = getAppointments();
  data[index].status = "Cancelled";
  data[index].bill = 0;

  saveAppointments(data);
  render();

  alert("Appointment cancelled.");
}

function clearAllAppointments() {
  if (confirm("Are you sure you want to clear all appointments?")) {
    localStorage.removeItem("appointments");
    render();
  }
}

render();
```

doctors.txt

```
D001|Dr. Rahman|Cardiologist
D002|Dr. Karim|Dermatologist
D003|Dr. Hasan|Neurologist
D004|Dr. Akter|Pediatrician
```